

MEMBRANE MECHANICS GOVERN DIPOLAR-ORDER RELAXATION IN MYELIN-LIKE MEMBRANES: IMPLICATIONS FOR MYELIN MRI

Niklas WALLSTEIN^{1*}, Axelle GRÉLARD², Olivier M. GIRARD^{1,3}, Seamoon DEB²,
Guillaume DUHAMEL^{1,3}, Antoine LOQUET^{2*} and Erick J. DUFOURC^{2*}

¹ Aix Marseille Univ, CNRS, CRMBM, Marseille, France

² University of Bordeaux, CNRS, Bordeaux INP, CBMN, UMR, 5248, IECB, Pessac, France

³ APHM, Hôpital Universitaire Timone, CEMEREM, Marseille, France

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Myelin integrity is essential for fast nerve conduction, yet the physical origin of myelin-sensitive MRI contrasts remains incompletely understood. In particular, the relationship between membrane mechanics and dipolar-order-based MRI signals, such as inhomogeneous magnetization transfer (ihMT), remains largely unknown. Here, we demonstrate that proton dipolar-order relaxation (T_{1D}) in myelin-like multilamellar membranes provides a direct readout of membrane mechanical dynamics. Using solid-state NMR measurements of systematically varied lipid-protein assemblies, we show that dipolar-order relaxation is governed by two distinct motional regimes: slow collective membrane undulations and faster molecular motions. At physiological temperatures, relaxation is dominated by collective undulations, identifying these motions as the principal determinants of T_{1D} and establishing a direct relationship between dipolar-order relaxation and membrane elasticity. Membrane composition systematically remodels this dynamical landscape. Cholesterol increases membrane rigidity and modifies both activation energies and correlation times, cerebroside further enhance membrane ordering while accelerating dipolar relaxation, whereas myelin basic protein dampens collective fluctuations and prolongs dipolar-order relaxation. These effects are consistently reflected in both dipolar relaxation times and spectral second moments. Together, our findings establish a mechanistic framework connecting membrane mechanics to dipolar-order relaxation in myelin-like membranes. More broadly, these insights provide a physical framework for interpreting dipolar-order-based MRI contrasts, including ihMT, as reporters of both the structural organization and the collective mechanical dynamics of myelin membranes.

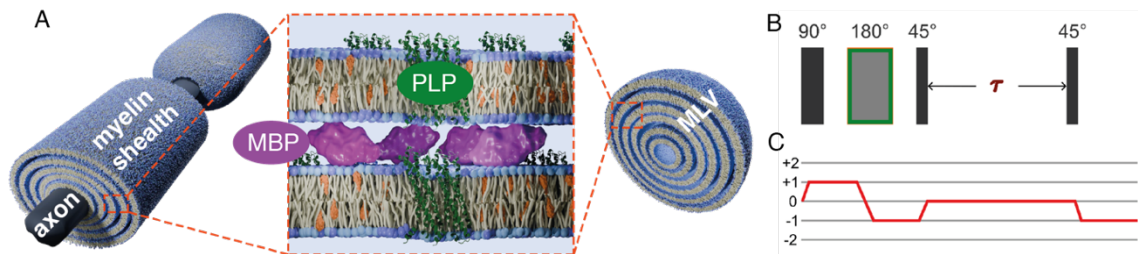


Figure 1. (A) Schematic representation of a myelinated axon (left) and a multilamellar vesicle (MLV, right). Phosphatidylcholine (PC) lipids, cholesterol and cerebroside represent the major lipids whereas myelin basic protein (MBP) and myelin proteolipid protein (PLP) represent the major proteins. Pulse sequence diagram (B) and coherence transfer map (C) of the chemical-shift-refocused Jeener-Broekaert sequence (red line indicates one possible pathway) to measure T_{1D} and pure proton dipolar spectra.